



Review Article

School-based environment and physical activity in adolescents: A systematic review and meta-analysis

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ABSTRACT

Objective: This study aimed to evaluate the effectiveness of school environmental factors in promoting adolescents' physical activity.

Methods: A systematic search of five databases (PubMed, Embase, Web of Science, EBSCO, and Cochrane Library) was conducted from the earliest available records up to September 2023. Meta-analyses were performed for each school environmental factor, provided that at least two studies reported on the association between that factor and adolescents' physical activity. The factors examined included seven school social environmental factors and six school physical environmental factors.

Results: 18 studies comprising 340,749 participants were included in the meta-analysis. A significant positive correlation was observed between the school environment and adolescents' physical activity. Specifically, the school social environment was positively associated with adolescents' physical activity. In particular, school-offered daily physical education, school-organized extracurricular physical activities, encouragement of using sports equipment or facilities, and outdoor obstacle course were all significantly positively correlated with adolescents' physical activity. While the overall school physical environment did not show a significant relationship with adolescents' physical activity, the number of school physical activity facilities demonstrated a significant positive correlation with adolescents' physical activity.

Conclusions: This meta-analysis demonstrates the role of school environment in promoting adolescents' physical activity. School social environmental factors, including school-offered daily physical education, school-organized extracurricular physical activities, encouragement to use sports resources, and outdoor obstacle courses, are particularly effective in enhancing adolescents' physical activity. Among school physical environmental factors, the number of physical activity facilities was found to be a key factor in promoting adolescents' physical activity.

1. Introduction

The health benefits of a physically active lifestyle during adolescence are well-established. Regular participation in a variety of organized physical activities is associated with improved adolescent well-being, including enhanced physical fitness (Rodriguez-Ayllon et al., 2019), cardiorespiratory endurance, bone health, cardiometabolic function (Poitras et al., 2016), attention, learning ability, and even academic

performance (Biddle et al., 2019). Several studies have shown that the physical activity developed during adolescence tends to track into adulthood (Tapia-Serrano et al., 2023; Yu et al., 2023). Therefore, promoting physical activity during adolescence is crucial for establishing lifelong healthy habits and preventing non-communicable diseases, such as obesity (Yu et al., 2023), diabetes, hypertension (Dou et al., 2023), and chronic musculoskeletal pain (Nascimento Leite et al., 2023). The World Health Organization recommends adolescents engage in an

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average of 60 min per day of moderate-to-vigorous intensity physical activity (MVPA) over the course of a week to derive health benefits (World Health Organization, 2020). However, a recent global estimate showed that more than three-quarters (1.6 million, 81 %) of adolescents aged 11–17 years from 146 countries fail to meet the recommendation (Guthold et al., 2020). As a result, the gap between the recommended and actual physical activity levels among adolescents has prompted public health policies to prioritize the promotion of physical activity during adolescence (Ferreira et al., 2007).

Adolescents, as a group of school-aged individuals, spend approximately four to eight hours at school each day for 12 years or more (Carson et al., 2014; De Meester et al., 2014). Schools serve as a key environment for adolescents to engage in physical activity. Given its educational responsibilities, schools also play a vital role in promoting adolescents' health and safety and in guiding the establishment of life-long healthy behaviors (ASCD and Centers for Disease Control and Prevention, 2018). Schools provide multiple opportunities for promoting adolescents' physical activity, including physical education classes, extracurricular activities before and after school, class breaks, active school commuting, and classroom-based physical activity (Singh et al., 2017; Watson et al., 2017). These opportunities can enable adolescents to meet the recommended 60 min of physical activity per day (Institute of Medicine, 2013; Piercy et al., 2015).

The motivation for adolescents' physical activity is often influenced by school-based environmental factors (Wang et al., 2017). Human behavior, especially among adolescents who may have limited conscious reflection, is frequently automatic and shaped by environmental stimuli rather than by conscious consideration of the consequences (Marteau et al., 2012). School-based environmental factors affecting physical activity can be categorized into physical aspects (e.g., physical structures and facilities) and social aspects (e.g., school support, physical activity promotion policies, and school norms). However, there is no consensus on whether the school physical environment promotes adolescents' physical activity. For instance, a cross-sectional study involving 14,077 American adolescents found that students in high school with a larger number of sports facilities were more likely to engage in MVPA (Hood et al., 2014). Conversely, Ward et al. and Kopcakova et al. found no correlation between the number of physical activity facilities in schools and improvements in adolescents' physical activity (Kopcakova et al., 2018; Ward et al., 2015). Similarly, the impact of the school social environment on adolescents' physical activity is also unclear. A study including 776 primary school students from 16 Canada schools found that physical education was not significantly correlated with adolescents' physical activity (Ward et al., 2015). However, Mooses et al. demonstrated that students engaged in 12.8 min more MVPA and reduced sedentary behavior by 9.7 min on days when physical education was offered, compared to days without physical education (Mooses et al., 2017). Overall, the influence of school-based environmental factors on adolescents' physical activity remains controversial.

Previous systematic reviews have primarily focused on evaluating the effectiveness of school-based interventions aimed at promoting physical activity among adolescents (Lee et al., 2023; Neil-Sztramko et al., 2021, p. 18; Parrish et al., 2020). These reviews have assessed various approaches, including increasing knowledge through health education (Lee et al., 2023), improving the quality of physical education to enhance the intensity or duration of students' physical activity (Neil-Sztramko et al., 2021, p. 18), or implementing school recess activity interventions (Parrish et al., 2020). However, these studies have generally concentrated on the implementation of interventions, often neglecting the broader influence of the school environment itself (e.g., the availability of sports facilities or the presence of physical activity policies) on adolescents' physical activity. Therefore, schools should not merely be seen as implementers of physical activity interventions, but as comprehensive ecological systems that shape students' opportunities and motivation to engage in physical activity. All components of the school environment should contribute to fostering adolescents' physical

activity.

Therefore, this study aims to conduct a systematic review and meta-analysis to compare the effect sizes of various school-based environmental factors (i.e. school physical and social environmental factors) on adolescents' physical activity.

2. Methods

2.1. Search strategy and study selection

This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (Page et al., 2021) (Supplementary material, Table S.1). Given that this study is based on published open data and does not involve new human trials or private data, ethics committee approval was not required. The protocol was registered on the International Prospective Register of Systematic Reviews, registration number: CRD42023470492. No similar systematic review protocol currently exists.

Five databases (PubMed, Embase, Web of Science, EBSCO, and Cochrane Library) were searched using the following keywords through Truncation and Boolean logic searching: ("adolescen*" OR "student*" OR "teen*" OR "youth") AND ("physical activity") AND ("school environment" OR "school social environment" OR "school built environment" OR "school physical environment*" OR "school equipment*" OR "school building*" OR "schoolyard*"). The search terms and strategy are provided in supplementary Table S.2. The peer-reviewed studies published in academic journals up to September 21, 2023, were included. Two authors (H.-H.W. and J.P.) independently screened and extracted relevant studies by reviewing titles, abstracts, and full-text. Any disagreement was resolved through discussion or by consulting a third arbitrator (X.H.).

2.2. Inclusion criteria

Studies were considered eligible for inclusion if: (1) participants were adolescents aged 10 to 19 years (World Health Organization, 2023); (2) the school internal environmental factors were analyzed as the independent variable; (3) the adolescents' physical activity (e.g., MVPA time, total physical activity time, or total physical activity-induced energy consumption) was identified as the dependent variable; (4) provided or allowed for the calculation of odds ratios (ORs) and 95 % confidence intervals (CIs) for the association between school environmental factors and adolescents' physical activity; and (5) the peer-reviewed studies published in English.

2.3. Exclusion criteria

Studies were excluded if they met any of the following criteria: (1) reviews, abstracts, case reports, non-peer-reviewed studies (e.g., dissertations or conference posters); (2) participants diagnosed with mental or orthopedic/neurological disorders that limited their ability to engage in physical activity; (3) the specific school environmental factors could not be quantitatively measured; and (4) incomplete data or inadequate descriptions of research methods in the study.

2.4. Quality assessment

Two independent authors assessed the quality of the selected studies using the Agency for Healthcare Research & Quality criteria, which includes 11 items with responses categorized as "yes," "no," or "unclear" (Rostom et al., 2004). If a study answered "no" or "unclear" to a given item, the for that item was "0", and if the answer was "yes", the score was "1". The total score was calculated to evaluate the study's quality: 0–3 indicates low quality, 4–7 indicates medium quality, and 8–11 indicates high quality. It is important to note that the quality assessment was not used to exclude studies from the review. Any disagreements

were resolved through discussion or by consulting a third arbitrator (X. H.).

2.5. Data extraction

Two authors independently extracted relevant data from the eligible studies using piloted data extraction sheets. Extracted information included authors, publication year, country/region, participant characteristics (e.g., age and gender), sample size, data measurement (i.e., school environment measurements and physical activity measurements), school environmental factors, and adjustment variables. When multiple effect sizes were reported for a correlation (e.g., multiple school physical activity promotion policies on physical activity), the overall effect size was used for the meta-analysis. Meta-analyses were performed for different school environmental factors, providing that at least two studies reported results on the association between the same school environmental factor and adolescents' physical activity. Based on previous research (Button et al., 2013; Prins et al., 2012; Schulz et al., 2005), school environmental factors were categorized into two groups: school social environmental factors and school physical environmental factors. School social environmental factors encompass policy- and interaction-related aspects, such as the provision of physical activity-related courses, the organization of extracurricular physical activities, physical activity promotion policies, and institutional support for physical activity. School physical environmental factors pertain to the natural environment, the quantity of sports facilities, and the built environment within the school.

If studies presented outcomes in figures rather than specific value, the GetData Graph Digitizer 2.26 would be used to extract the necessary data. If the OR and 95 % CI were not presented in the studies, we would contact the authors by email. Studies were excluded if the authors did not respond.

2.6. Statistical analyses

The adjusted (if available) or non-adjusted (if adjusted data were not reported) ORs with 95 % CIs were used to assess the associations between school environmental factors and adolescents' physical activity. Heterogeneity among studies was calculated using the I^2 index. According to the thresholds proposed by Huedo-Medina et al., heterogeneity was classified as low ($I^2 \leq 25\%$), moderate ($25\% < I^2 \leq 50\%$), high ($50\% < I^2 \leq 75\%$), and very high ($I^2 > 75\%$) (Huedo-Medina et al., 2006). For analyses with low or moderate heterogeneity, a fixed-effects model was applied. In contrast, a random-effects model was used when heterogeneity was high or very high. Subgroup analyses were performed to explore potential sources of heterogeneity based on school environmental factors (e.g., school physical environment and school social environment), gender, school type (i.e., primary school and secondary school), and physical activity type (i.e., total physical activity, TPA and MVPA). Egger's test and Begg's test were employed to detect publication bias, with $P > 0.05$ indicating no significant bias. Sensitivity analyses were performed to assess the stability of the meta-analysis results. If excluding any study did not alter the overall effect size or its CI, the results were considered stable. All analyses were performed using Stata Statistical Software (version 16.0; StataCorp LLC, College Station, Texas).

3. Results

3.1. Search results

The flowchart in Fig. 1 presents the search procedure. An initial search of five databases identified 1557 records, with 1071 remaining after duplicate removal. 48 potentially eligible studies remained after screening titles and abstracts. Among them, 30 studies were excluded due to the following reasons: (1) the school neighborhood

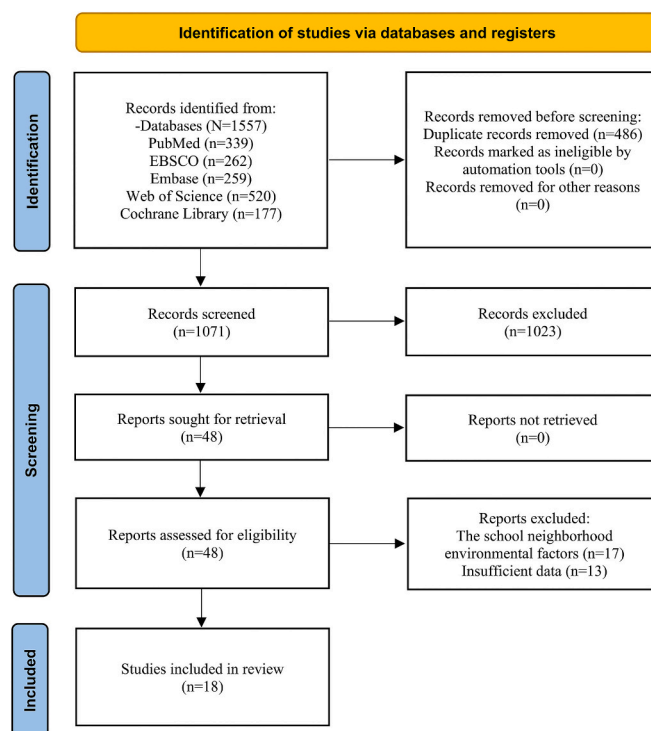


Fig. 1. The flowchart of the search procedure.

environmental factors ($n = 17$) and (2) insufficient data ($n = 13$). Finally, 18 studies comprising 68 data points (340,749 participants) met the inclusion criteria and were included in this meta-analysis.

3.2. Characteristics of included studies

Table 1 summarizes the characteristics of the included studies. Quality assessment results of all studies were demonstrated in supplementary Table S.3. Six studies were rated as high quality and 12 studies were rated as medium quality. The average quality score of the overall studies was 6.83. All studies involved adolescents aged 10–19 years, with 17 studies including both genders and one study focusing exclusively on girls. All studies were cross-sectional design, with school environmental factors assessed via questionnaires ($n = 14$, 78 %) or interviews ($n = 4$, 22 %). Of these 18 included studies, three studies (17 %) used accelerometers (Martin et al., 2014; Morton et al., 2016; Nyberg et al., 2021) and 15 studies (83 %) used questionnaires to measure adolescents' physical activity (Amornsriwatanakul et al., 2023; da Silva et al., 2023; de Rezende et al., 2015; Durant et al., 2009; Haug et al., 2010; Haug et al., 2008; Hobin et al., 2010; Hood et al., 2014; Kirby et al., 2012; Kopcakova et al., 2018; Morgan et al., 2016; Oliveira et al., 2018; Peralta et al., 2019; Perry et al., 2011; Ward et al., 2015), respectively.

Of these 18 studies, 13 studies investigated school social environmental factors ($n = 13$, 72 %) and 12 studies explored school physical environmental factors ($n = 12$, 67 %). A total of 13 types of school environmental factors were examined, including seven social factors and six physical factors. School social environmental factors included school-offered daily physical education, school-offered intramural sports, school-organized interschool sports, school-organized extracurricular physical activities, school physical activity promotion policies, the encouragement of using sports equipment or facilities, and outdoor obstacle course. School physical environmental factors included skating areas, school grass areas, boarding areas, ball activities areas, the number of school physical activity facilities, and swimming areas.

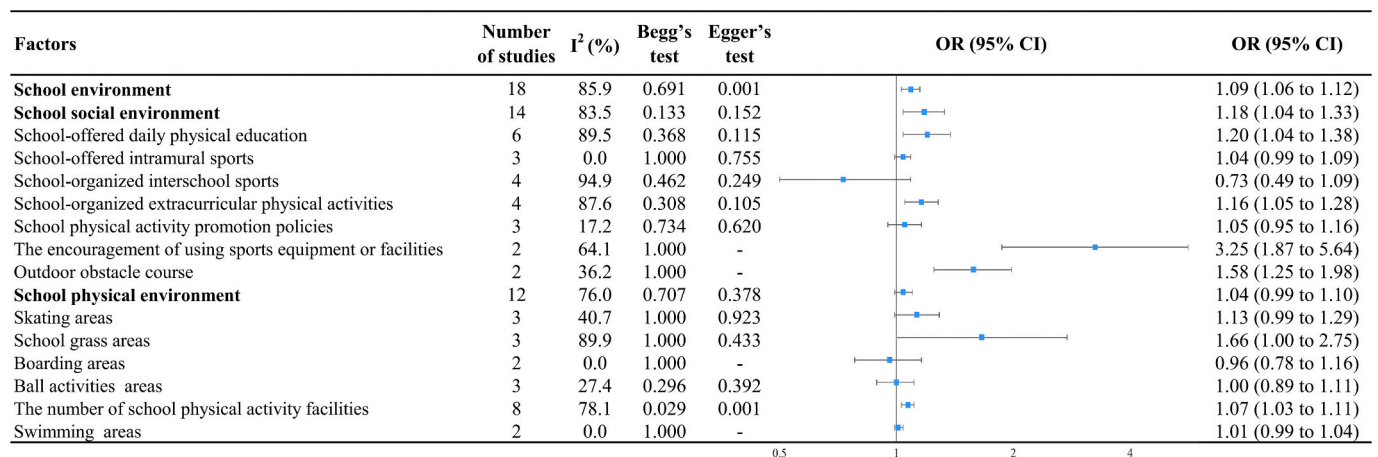
Table 1

Characteristics of included studies on school environmental factors and adolescents' physical activity, published from the earliest available records to September 2023.

No.	Authors (publication year)	Country	Sample size	Age (years)	Girl (%)	School environment measurements	Physical activity measurements	School environmental factors	Adjustments
1	Hobin et al. (2010)	Canada	2379	10-13	52.7	Interview	Questionnaire	B, C	Age
2	Kopcakova et al. (2018)	Slovakia	9743	13.5	49.7	Questionnaire	Questionnaire	H, L, K	Age, gender
3	Ward et al. (2015)	Canada	776	10-11	55.2	Questionnaire	Questionnaire	A, B, C, E, L	Age, socioeconomic status, and language
4	Kirby et al. (2012)	UK	1978	13.5	100.0	Questionnaire	Questionnaire	C, D	Age and family affluence scale
5	Nyberg et al. (2021)	Sweden	1139	13.4 ± 0.3	49.0	Questionnaire	Accelerometer	A	Gender, parental education, school clustering, and accelerometer wear time
6	Amornsriwatanakul et al. (2023)	Thailand	4617	14-17	50.3	Questionnaire	Questionnaire	I	Clustering
7	Peralta et al. (2019)	Australia	2317	12-17	49.9	Interview	Questionnaire	F	Students' language background and residence
8	Perry et al. (2011)	USA	773	10-19	51.0	Questionnaire	Questionnaire	A, D	Age and gender
9	da Silva et al. (2023)	Brazilian	81,154	14.3 ± 1.3	51.3	Questionnaire	Questionnaire	M	Age, gender, maternal education, and school area
10	Hood et al. (2014)	USA	14,077	14-17	–	Questionnaire	Questionnaire	L	Year, gender, race/ethnicity, parental education, living with both parents, school-level school size, geographic region, urbanicity, and free or reduced-price lunch eligibility
11	Haug et al. (2008)	Norway	1347	13	47.7	Questionnaire	Questionnaire	G, H, J, K, L, M	Socioeconomic status, gender, and individual interests in school physical activity
12	Martin et al. (2014)	Australia	408	11.1 ± 0.4	45.8	Questionnaire	Accelerometer	I	Total time children spent in class
13	Morgan et al. (2016)	UK	7376	13.7 ± 1.4	49.1	Questionnaire	Questionnaire	A, E, L	–
14	Durant et al. (2009)	USA	165	14.6 ± 1.7	53.0	Interview	Questionnaire	A, F	Race-ethnicity, site, gender, and age
15	Oliveira et al. (2018)	Brazilian	102,072	15	–	Questionnaire	Questionnaire	B, C, D, L	–
16	Morton et al. (2016a, 2016b)	UK	325	10.2 ± 0.3	52.3	Questionnaire	Accelerometer	A, E	Gender, age, BMI, and family socioeconomic status
17	Haug et al. (2010)	Norway	6858	13-15	50.0	Questionnaire	Questionnaire	G, H, I, J, K, L	–
18	de Rezende et al. (2015)	Brazilian	109,104	14-15	52.0	Interview	Questionnaire	D, L	Age, gender, mother's education level, region, municipality, type of school, and number of enrolled students in the school

A: School-offered daily physical education; B: School-offered intramural sports; C: School-organized interschool sports; D: School-organized extracurricular physical activities; E: School physical activity promotion policies; F: The encouragement of using sports equipment or facilities; G: Outdoor obstacle course; H: Skating areas; I: School grass areas; J: Boarding areas; K: Ball activities areas; L: The number of school physical activity facilities; M: Swimming areas.

BMI: Body Mass Index.

**Fig. 2.** Associations between school environmental factors and adolescents' physical activity, based on studies published from the earliest available records to September 2023.

Note: Analyses with $I^2 \leq 50\%$ were conducted using a fixed-effects model, while those with $I^2 > 50\%$ were based on a random-effects model. Begg's test and Egger's test were used to assess publication bias, with $P > 0.05$ indicating no significant bias.

3.3. Integrated associations between school environment and adolescents' physical activity

Fig. 2 shows the diversity of school environmental factors and their associations with adolescents' physical activity. The pooled analysis revealed a significant positive association between school environment and adolescents' physical activity (OR = 1.09; 95 % CI = 1.06 to 1.12; I^2 = 85.9 %). The associations between school environmental factors and physical activity were grouped by school social environments (i.e., school-offered daily physical education, school-offered intramural sports, school-organized interschool sports, school-organized extracurricular physical activities, school physical activity promotion policies, the encouragement of using sports equipment or facilities, and outdoor obstacle course) and school physical environments (i.e., skating areas, school grass areas, boarding areas, ball activities areas, the number of school physical activity facilities, and swimming areas).

3.3.1. Associations between school social environment and adolescents' physical activity

13 studies demonstrated the effectiveness of school social environment on adolescents' physical activity. As shown in Fig. 2, the school social environment was positively associated with adolescents' physical activity (OR = 1.18; 95 % CI = 1.04 to 1.33; I^2 = 83.5 %). Specifically, the school-offered daily physical education (OR = 1.20; 95 % CI = 1.04 to 1.38; I^2 = 89.5 %), school-organized extracurricular physical activities (OR = 1.16; 95 % CI = 1.05 to 1.28; I^2 = 87.6 %), the encouragement of using sports equipment or facilities (OR = 3.25; 95 % CI = 1.87 to 5.64; I^2 = 64.1 %), and outdoor obstacle course (OR = 1.58; 95 % CI = 1.25 to 1.98; I^2 = 36.2 %) were significantly associated with increased adolescents' physical activity.

Subgroup analyses by gender, school type, and physical activity type are shown in Fig. 3. For gender, there were no significant differences in the associations between adolescents' physical activity and the school social environment, including specific components such as school-offered daily physical education, school-organized interschool sports.

For school type, a significantly positive correlation was observed between the school social environment and adolescents' physical activity in secondary school (OR = 1.27; 95 % CI = 1.17 to 1.37; I^2 = 92.4 %). Moreover, school-offered daily physical education was significantly positively associated with adolescents' physical activity in both primary (OR = 1.01; 95 % CI = 1.01 to 1.02; I^2 = 0 %) and secondary schools (OR = 1.45; 95 % CI = 1.11 to 1.89; I^2 = 92.3 %). Regarding physical activity type, adolescents' TPA was positively associated with the school social environment (OR = 1.29; 95 % CI = 1.18 to 1.40; I^2 = 93.3 %) and school-organized extracurricular physical activities (OR = 1.14; 95 % CI = 1.04 to 1.25; I^2 = 90.2 %), respectively. Additionally, adolescents' MVPA was significantly positively associated with school-organized extracurricular physical activities (OR = 1.58; 95 % CI = 1.06 to 2.36), respectively.

3.3.2. Associations between school physical environment and adolescents' physical activity

12 studies demonstrated the associations between school physical environment and adolescents' physical activity. As shown in Fig. 2, the school physical environment was not significantly associated with adolescents' physical activity (OR = 1.04; 95 % CI = 0.99 to 1.10; I^2 = 76.0 %). However, the number of school physical activity facilities (OR = 1.07; 95 % CI = 1.03 to 1.11; I^2 = 78.1 %) was significantly associated with increased adolescents' physical activity.

Subgroup analyses by gender, school type, and physical activity type are shown in Fig. 3. Regarding gender, a significant positive correlation was observed between the school physical environment and adolescents' physical activity among girls (OR = 1.02; 95 % CI = 1.00 to 1.04; I^2 = 0 %). For school type, a significantly positive correlation was identified between school grass areas and adolescents' physical activity in primary school (OR = 3.32; 95 % CI = 2.01 to 5.48). In secondary schools, school physical environment (OR = 1.06; 95 % CI = 1.03 to 1.09; I^2 = 70.0 %) and the number of school physical activity facilities (OR = 1.06; 95 % CI = 1.02 to 1.09; I^2 = 79.9 %) were significantly positively associated with adolescents' physical activity. For physical

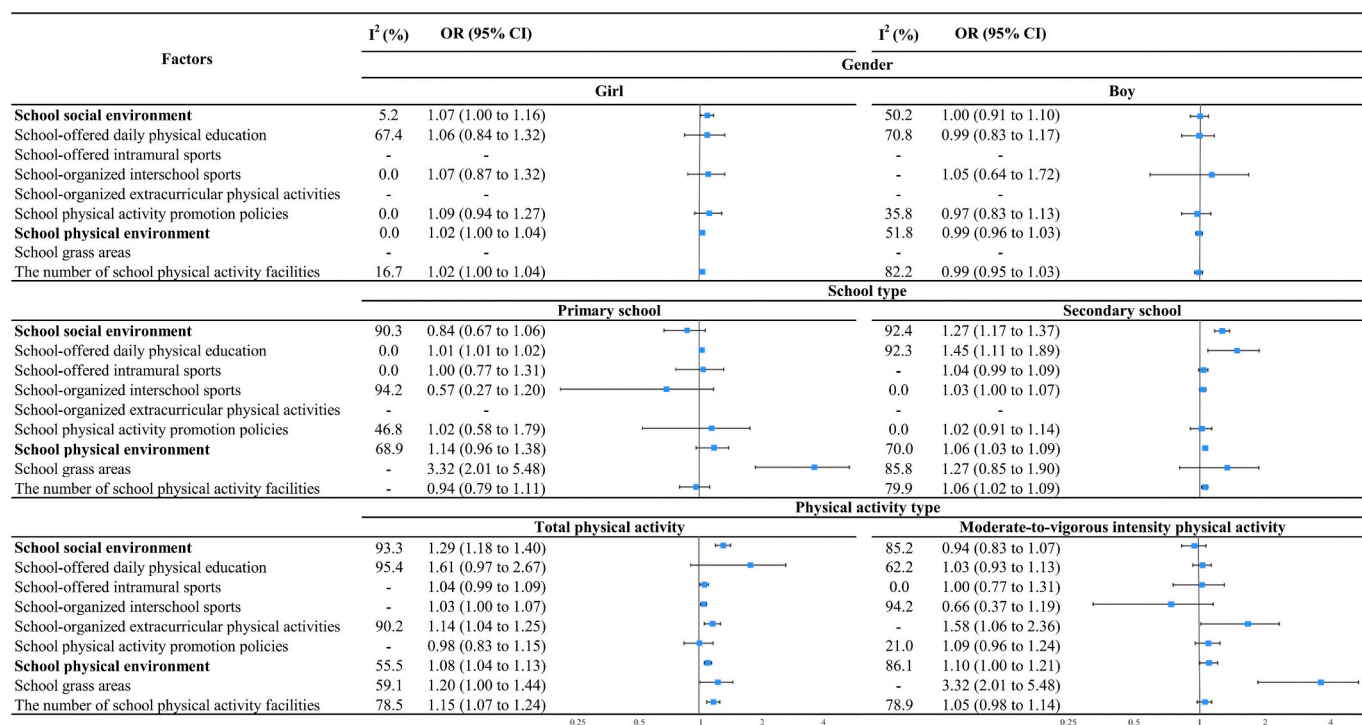


Fig. 3. Subgroup analyses of the association between school environmental factors and adolescents' physical activity, stratified by gender, school type, and physical activity type, based on studies published from the earliest available records to September 2023.

Note: Analyses with $I^2 \leq 50$ % were conducted using a fixed-effects model, while those with $I^2 > 50$ % were based on a random-effects model.

activity type, adolescents' TPA was positively associated with the school physical environment (OR = 1.08; 95 % CI = 1.04 to 1.13; $I^2 = 55.5\%$) and the number of school physical activity facilities (OR = 1.15; 95 % CI = 1.07 to 1.24; $I^2 = 78.5\%$), respectively. Additionally, school grass areas were significantly positively associated with adolescents' MVPA (OR = 3.32; 95 % CI = 2.01 to 5.48).

3.4. Publication bias and sensitivity analysis

Egger's test indicated publication bias for school environment ($P < 0.05$). Both Begg's test and Egger's test revealed bias for the number of school physical activity facilities ($P < 0.05$). However, sensitivity analyses demonstrated that the overall results for school environment and the number of school physical activity facilities remained stable, with no significant changes in effect sizes following the exclusion of any single study (Supplementary material, Fig. S.1, S.2).

4. Discussion

This meta-analysis synthesizes data from 18 studies examining the effects of school physical and social environments on adolescents' physical activity. To the best of our knowledge, this is the first study to evaluate the specific effect of various school environmental factors on promoting adolescents' physical activity. Our findings suggested that, in general, school-based physical activity-relevant environments can positively influence adolescents' physical activity. Specifically, the school social environment was significantly associated with increased adolescents' physical activity, while the school physical environment was not significantly associated with adolescents' physical activity.

Among the components of the school social environment, school-offered daily physical education, outdoor obstacle course, school-based extracurricular physical activities, and encouragement of using sports equipment or facilities were particularly effective in promoting adolescents' physical activity. School-offered daily physical education creates structured opportunities for students to be active without detracting from academic time. It enables students to engage in a variety of physical activity while simultaneously learning physical, social, mental, and emotional skills (Wiium, 2021). Additionally, physical education positively affects certain aspects of mood, such as increased energy, relaxation, and enjoyment, while also decreasing exhaustion, tension, and depression, which probably contributes to increased MVPA time for students (Pasek et al., 2022). The outdoor obstacle course, which incorporates elements of challenge and competition, stimulates adolescents' active participation in physical activity. These courses typically involve various activities, such as running, jumping, climbing, and crawling, all of which promote overall physical fitness and skill development (Haug et al., 2010; Haug et al., 2008). School-organized extracurricular physical activities offer additional opportunities for adolescents to engage in physical activity beyond regular physical education (Mears and Jago, 2016). These activities may offer a diverse range of sports, sparking adolescents' interest in exercise and encouraging greater participation in physical activity. According to Self-Determination Theory, supportive behaviors from significant others (e.g., teachers) can shape adolescents' motivation toward a particular activity by satisfying their psychological needs (Ryan and Deci, 2017). Hence, encouragement from teachers to use sports equipment or facilities can enhance adolescents' motivation to collaborate and try new activities (Perry et al., 2011).

In terms of the school physical environments, only the number of school physical activity facilities significantly promoted adolescents' physical activity. Physical activity facilities, such as basketball hoops, table tennis tables, nets, and dumbbells, serve as external stimuli encouraging adolescents to participate in physical activity (de Rezende et al., 2015; Haug et al., 2008; Hood et al., 2014). The greater the number of physical activity facilities at a school, the stronger the external stimulus for physical activity participation. For instance, each

additional physical activity facility (e.g., nets, soccer and goals, long-jump pits, and wall markings) was associated with an additional five minutes of daily MVPA per adolescent (Martin et al., 2014). Sufficient physical activity facilities provide students with more opportunities to engage in physical activity and reduce competition for facility access.

Subgroup analyses in this study indicated that the association between school environments and adolescents' physical activity varies by school types. The school social and physical environments of secondary schools were significantly and positively associated with adolescents' physical activity, whereas no such association was observed in primary schools. This disparity may be attributed to the larger size of secondary schools, which typically boast better-equipped sports facilities and more opportunities for physical activity, fostering greater participation among students (Harrison et al., 2016). Furthermore, because primary school students are younger, teachers may closely supervise and guide them for safety, which could limit their independent engagement in physical activity (McGrath et al., 2015).

This study also found that different school social and physical environmental factors exert varying degrees of influence on adolescents' physical activity. Rather than being driven by a single environmental factor, adolescents' physical activity is influenced by the combination and interaction of diverse school environmental factors. Potential mediating relationships may exist among the school physical environment, the school social environment, and adolescents' physical activity (Eather et al., 2013). For instance, while the school physical environment alone may not be sufficient to promote physical activity, it serves as a critical facilitator of physical activity behaviors (Giles-Corti and Donovan, 2003). Features of the school physical environment, such as the presence of sports areas and facilities, provide essential opportunities for engaging in physical activity. However, the school social environment plays a pivotal role in encouraging and facilitating the use of these resources (McCormack et al., 2010). Without active encouragement and support from the broader school social environment, the potential benefits of physical resources may remain underutilized, leading to a negligible overall impact on physical activity (Morton et al., 2016a). Future research should focus on integrating physical and social environmental factors to examine how these elements interact to influence adolescents' physical activity. This would help identify causal pathways for promoting physical activity among adolescents and provide guidance for effective school-based interventions.

Although this is the first meta-analysis to evaluate the effectiveness of various school environmental factors on promoting adolescents' physical activity, several limitations must be acknowledged. First, most of the included studies used subjective methods to measure physical activity and school environments, which could introduce information bias, as participants may be inclined to avoid negative responses (Knuth and Hallal, 2012). Second, the number of studies included in this meta-analysis was limited ($n = 18$), especially for certain specific school environmental factors. Therefore, the evidence for the associations between school environmental factors and adolescents' physical activity may need to be re-examined once more original research is available. Finally, heterogeneity, publication bias, low-quality studies, and language limitations of the included studies may have non-objectively influenced the estimated effect sizes. Differences between studies may be partly explained by the socio-cultural variations in samples, differences in physical activity measurement methods, variations in school environment measures, and differences in physical activity intensity thresholds. Despite our efforts to control for potential sources of bias in this meta-analysis, we were unable to entirely avoid these limitations.

5. Conclusion

This systematic review and meta-analysis demonstrates that the school-based environment plays a crucial role in promoting adolescents' physical activity. Among various school social environmental factors, school-offered daily physical education, school-organized

extracurricular physical activities, the encouragement of using sports equipment or facilities, and outdoor obstacle courses are particularly effective in increasing adolescents' physical activity. Among school physical environmental factors, the number of physical activity facilities was found to be a key factor in promoting physical activity among adolescents.

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CRediT authorship contribution statement

Huihui Wang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Jian Pang:** Validation, Resources, Project administration, Methodology, Investigation. **Xiaoguang Yang:** Validation, Formal analysis, Data curation. **Yuanyuan Jia:** Writing – review & editing, Resources, Investigation. **Xinru Huang:** Writing – review & editing, Investigation. **Laikang Yu:** Supervision, Software, Data curation, Conceptualization. **Xiao Hou:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. All authors agree to the final version of the article and are responsible for its content.

Data availability

The datasets used and analyzed during the current study available from the corresponding author on reasonable request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ypmed.2025.108221>.

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